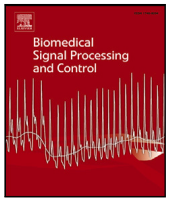
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Automatic diagnosis and localization of myocardial infarction using morphological features of ECG signal

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ABSTRACT

Electrocardiogram (ECG) is a non-invasive and economical diagnostic tool for detecting myocardial infarction (MI). The occurrence of a heart attack causes distortions in the ECG waves. This article extracts morphological features from ECG signals to detect and localize MI. After preprocessing the ECG signal, its fiducial points are identified. Then morphological features such as the amplitude, interval, and angle between waves are extracted. A random forest classifier with 100 trees has been used for classification and feature selection. The method was evaluated using the PTB dataset, containing 52 healthy and 148 MI subjects. We tried to diagnose and localize MI in two schemes: interpatient and intrapatient. In this method, we obtained superior results with an accuracy of 80.98%, a sensitivity of 80.98%, a specificity of 96.32%, a positive predictive value of 79.72%, and an F-score of 79.53% for MI localization in the interpatient scheme compared to the state-of-the-art. Our model achieves an accuracy of 96.54%, a sensitivity of 99.74%, a positive predictive value of 96.09%, and an F-score of 97.88% in the interpatient scheme detection. In the interpatient domain, 96.68% accuracy was obtained using only 6 chest leads for detection. The proposed method is interpretable with low computational complexity and applies a new package of morphological features. Compared to recent studies, in this study, the results have been improved in the interpatient scheme which has more vital clinical significance.

1. Introduction

Myocardial infarction (MI) is one of the most common cardiovascular diseases caused by coronary artery occlusion. This blockage severely reduces blood flow, damages heart tissue, and leads to a heart attack [1]. Failure to early treatment results in tissue death. According to the World Health Organization, 17.9 million people die each year from cardiovascular diseases (CVD), an estimated 32% of all deaths worldwide [2].

ECG is an efficient, non-invasive, and cost-effective tool for MI detection. Damage to the heart tissue is reflected on the ECG waveform. Based on the location of the injury, physiological distortions occur in 12 leads ECG, including pathologic Q wave, ST-segment elevation and depression, and T wave inversion [3]. These distortions can be leveraged to identify different types of MI.

Detection of changes in a heartbeat waveform and correlation between ECG leads have become the solution to automatic MI detection and localization.

Training a model that considers individual differences between people is important from a clinical point of view and is more practical in clinical practice. In the interpatient domain, the algorithm is trained using the data of some patients who are not present in the testing. On the other hand, in the intrapatient scheme, all records are separated for training and testing after the shuffle, and testing may have received a record of that patient in training.

Various methods have been proposed to detect and localize MI. These methods usually include feature extraction, feature dimensionality reduction, and classification. Feature extraction methods include time domain analyzes [4,5], frequency domain analyzes [6–10], optimization algorithms [11,12], entropy features [13–17], deep learning theory [18–20], HMM theory [21]. Locality Preserving Projections (LPP) [8], Principal Component Analysis (PCA) [22], and swarm intelligence algorithm [23] are the most common methods of reducing feature dimensions among these articles. Support Vector Machine (SVM) [5,9,15,17], K-Nearest Neighbor (KNN) [13,16], Decision Tree [23], simple Bayes [24], BackPropagation Neural Network (BPNN) [12] and other machine learning algorithms have been widely used as classification methods. Also, in recent years, deep learning architectures such as Convolutional Neural Network (CNN), Recurrent

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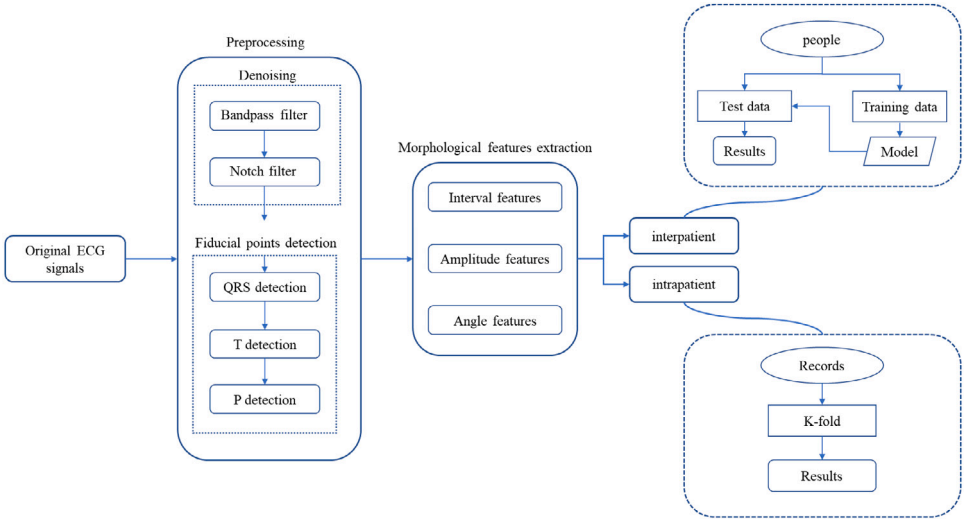


Fig. 1. Overview of the proposed method.

Neural Network (RNN), Restricted Boltzmann Machine (RBM), attention network, and autoencoder have been applied to physiological signals such as ECG [20,25–29].

There were some drawbacks in previous studies, despite the good results. Firstly, the performance needs to be improved for MI localization in the interpatient scheme. Secondly, although wavelet transform has been used in many studies to extract beneficial features from the frequency domain, methods based on wavelet transform are highly dependent on selecting basis functions. Thirdly, neural network approaches require a large amount of data to train the network model [30]. Fourthly, in CNN models, the feature extraction and the classification are integrated, so they lack interpretability. Finally, although there are many articles on the intrapatient scheme, few articles have considered the interpatient scheme, which is of great clinical importance.

This study used a new package of morphological features in MI detection and localization, such as interval, amplitude, and angle between different waves of each beat in the ECG signal. Morphological features are accurate indicators of the status of the understudy signals. The difference between the types of classes well and achieve better results in the interpatients domain with the same volume of data and database used in the literature. The main innovations of our proposed method are as follows:

- This algorithm applies a new package of morphological features and multi-lead feature fusion.
- It can investigate the impact of different leads and morphological changes in ECG for MI localization.
- The automatic diagnostic method has interpretability according to the medical strategy.
- This method has more vital clinical significance because, in addition to intrapatient, the interpatient scheme is performed.
- Our method has achieved superior results in MI localization of the interpatient scheme compared to recent studies.

This paper is organized as follows: Section 2 presents the proposed method. Section 3 presents the results. Section 4 discusses and compares our results to other related studies. Finally, Section 5 concludes the paper.

2. Materials and methods

Our study tends to extract information from 12 leads ECG using morphological features, efficiently classifying the pathological distortions caused by MI. Fig. 1 presents the block diagram of the proposed method. Details of each block are described in the following sections.

Table 1
Number of patients and records in each type of MI.

Class	Number of subjects	Number of records
HC	52	80
AMI	17	47
ASMI	27	77
ALMI	16	43
IMI	30	89
ILMI	23	56
Total	165	392

2.1. ECG database

The database used in this study is the PTB database [31,32], collected by the Cardiovascular Department of the Benjamin Franklin University Clinic in Berlin and provided by the German National Metrology Institute for research purposes. This database contains 549 records of 290 people. PTB database contains approximately two minute records of individuals. The data of each person is represented by one to five records. Each record contains 15 simultaneously measured signals: 12 regular leads (i, ii, iii, avr, avl, avf, v1, v2, v3, v4, v5, v6) with 3 Franks signals (vx, vy, vz). Each signal is sampled at 1000 Hz with a resolution of 16 bits in the range of 16.384 mV. The database does not have a beat-by-beat annotation. PTB includes 148 MI and 52 healthy control (HC). There are 11 types of MI which are as follows: anterior MI (AMI), anterior lateral MI (ALMI), anterior septal MI (ASMI), anterior septal lateral MI (ASLMI), inferior MI (IMI), inferior lateral MI (ILMI), inferior posterior MI (IPMI), inferior posterior lateral MI (IPLMI), lateral MI (LMI), posterior MI (PMI) and posterior lateral MI (PLMI).

Five types of MI are used in this study as there are enough patients in each type. These are anterior, anterior lateral, anterior septal, inferior, and inferior lateral. Table 1 shows the details of the number of people and records related to each type.

2.2. Preprocessing

The ECG signals are contaminated with low-frequency and high-frequency noises; thus, the first step is removing the noise through frequency filters. Then, the fiducial points of ECG are identified.

2.2.1. Denoising ECG signals

In this study, we used a 3rd order Butterworth filter with cut-off frequencies of 1 Hz and 100 Hz to remove high-frequency and low-frequency noises such as electromyographic noise and baseline

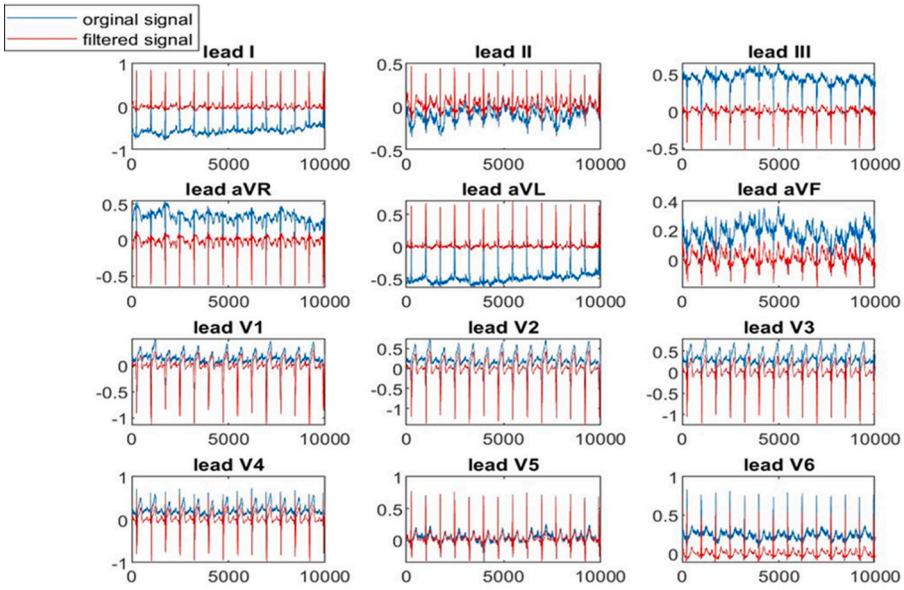


Fig. 2. Results for the noise removal of 12-lead ECG of s0020bre.

wandering. Then, a 50 Hz notch filter was used to remove the power line interference. The noise removal results on 12 lead ECGs from the record s0020bre are shown in Fig. 2.

2.2.2. Fiducial points detection

Fiducial Points (FPs) in an ECG signal are the location of the peak, onset, and offset of waveforms. After denoising, the ECG signal, the QRS complex, P, and T waves are detected. Each of these waves indicates electrical activity in different areas of the heart. Therefore, their waveform and amplitude can be used to diagnose heart disease. In [33], a new method was presented to detect waves accurately for all 12-leads of ECG. Its description is as follows:

QRS detection Fig. 3 shows the flowchart of QRS detection algorithm. At first, the signal is passed through a median filter with cut-off frequencies of 5 Hz and 100 Hz to attenuate T and P waves to get a QRS complex with prominent S and Q waves (see Fig. 4). A free phase stationary wavelet transform (SWT) is applied with the Haar function [34,35]. The SWD_N, which are detail coefficients of the SWT, is calculated at a frequency of about 45 Hz, and spectral components of the QRS complex can be detected at this frequency span [36]. The SWD of N+1 levels of record s0036lre, which should be decomposed to level 3 (N = 3), is given in Fig. 5. A threshold-based method for detecting intervals with QRS complex is considered. This threshold is determined according to the variable time windows applied to SWD_N and the statistical measurement of the signal [33]. The position with the longest amplitude in the QRS complex is identified as the R_{peak}. The Q and the S waves are minimum amplitude between the start of the QRS complex to the R_{peak} and between the R_{peak} to the end of the QRS complex. The QRS_{onset} and QRS_{offset} are 20 ms before Q_{peak} and after S_{peak}, respectively.

T wave detection This method depends on QRS detection, which is done for each lead individually. Preprocessing steps such as QRS replacement, filtering, and SWT are required for T wave detection. At first, a sigmoid function replaces a constant segment at each RR interval so that the QRS complex and the P wave are almost flattened. Then, a 4th order Butterworth bandpass filter with cut-off frequencies of 0.3 Hz and 20 Hz is applied to the signal. This QRS-free signal enters the free phase SWT [35] using the rbio3.3 function [37]. In this case, the T wave center frequency, 7 Hz, determines the level N of the detail coefficients [36]. Then, the T wave polarity is estimated by considering all the extrema in the SWT signal. The T_{peak} at each RR interval is considered by selecting the maximum or minimum amplitude for positive or negative polarity in a search window that relies on the RR interval and the QRS_{offset}. Considering the polarity, the T wave boundaries are determined in the time domain. The area beneath the SWT curve corresponding to level N-1 is calculated for each T wave in the RR interval. The T_{onset} and T_{offset} are assessed according to the two thresholds. These two thresholds are intended for the distribution of the T wave region and the position of the other peaks in the interval.

P wave detection It also depends on QRS and T wave detection. The signal corresponding to one lead enters the Gaussian bandpass filter with cut-off frequencies of 1 to 15 Hz. The sigmoid function replaces the QRS complex and the T wave. SWT is accomplished using a quadratic spline wavelet with N-level detail coefficients according to the center frequency of the P wave, in this case, 7 Hz [36]. In each RR interval and according to the position of the Q wave, the P wave is determined by annotating the SWT peaks absolute maximum. Next, the P_{onset} and P_{offset} are specified. Due to the low P wave amplitude, a SWT pattern is constructed in a window of 50 waves around the P_{peak} for boundary estimation. Hotelling's T-squared is used to catch a clear pattern. The P_{onset} and P_{offset} are appraised by considering the initial guesses according to the closest wavelet extremum.

2.3 Feature extraction

MI in ECG becomes evident as morphological changes, such as Hyperacute T wave, ST-segment elevation or depression, and pathological Q wave.

The set of features used in this study is related to the intervals between the peaks and the waves in an ECG beat. The time interval features have been explained in Table 2 in detail. Also, they are shown in Fig. 6.

The amplitude features are calculated by considering the R_{peak}. RQ_A, RS_A, RP_A, and RT_A are the amplitude differences between R_{peak} and Q_{peak}, S_{peak}, P_{peak}, and T_{peak}, respectively. Also, you can see an overview of these features in Fig. 6.

Angle features are the angle Q, the angle R, and the angle S. Angle Q is the angular displacement between the directed lines that connect the Q wave to the R_{peak} and P_{peak}. Similarly, the angle R is the angular displacement between the directed lines that link the R_{peak} to the Q wave and the S wave. Also, the angle S is the angular displacement between the directed lines that join the S wave to the T_{peak} and the

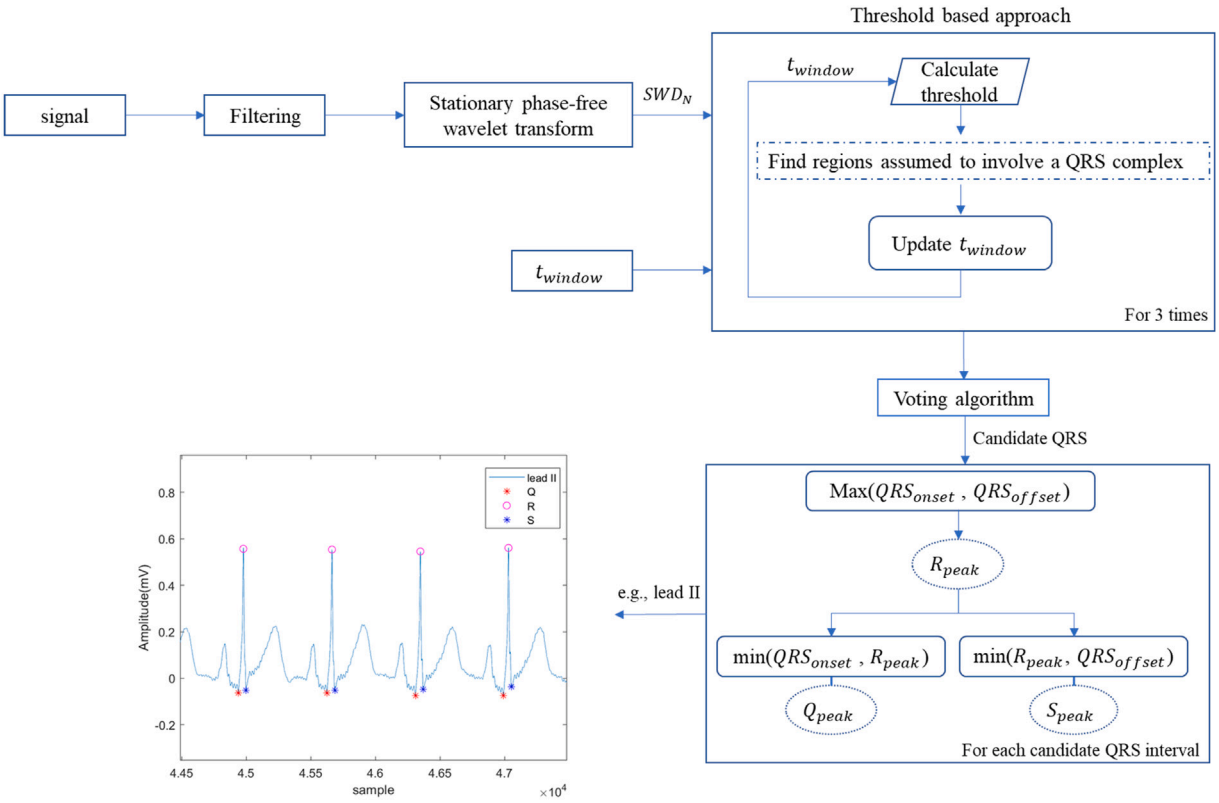


Fig. 3. Flowchart of QRS detection algorithm. SWD_N: detail coefficients at Nth level of SWT, t_window: length of search window.

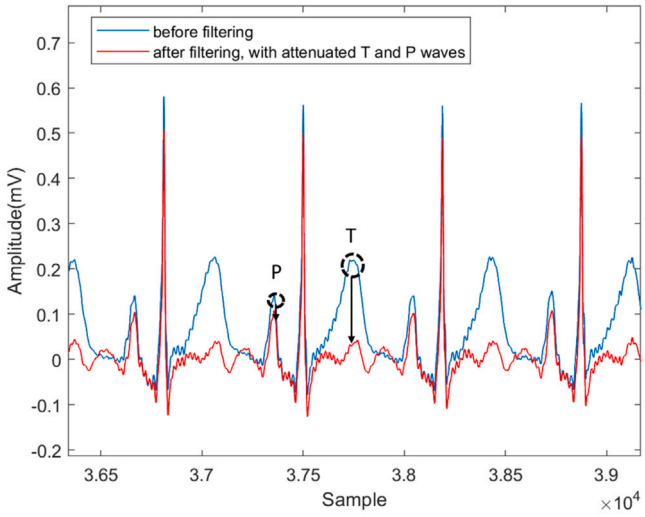


Fig. 4. Signal before and after median filter with attenuated T and P waves.

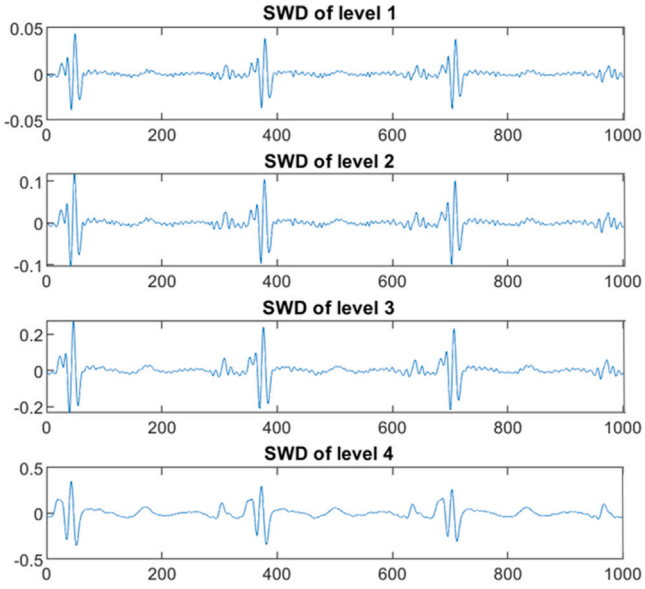


Fig. 5. Detail coefficients of four levels of record s0036lre.

R_{peak}. Angle features are defined as shown in Fig. 7. Angles have been calculated through the four-quadrant inverse tangent of the norm of the cross product and the dot product of x and y, which are the vectors obtained by subtracting the positions of peaks.

2.4 Constructing a TreeBagger for MI classification and feature selection

Random forest is defined as the bagging applied to the decision trees [38]. This classifier is a type of ensemble learning. Bagging is known as Bootstrap aggregating. To create a bootstrapped dataset that is the same size as the original, we randomly select samples from the original dataset. The important detail is that the same sample can be picked more than once. The bootstrapped dataset is created from the original data. Decision trees are applied to each Bootstrap. In our work, N = 100 is appropriate because it minimizes the Out-Of-Bag (OOB) error in the TreeBagger classifier. One decision tree that is made using the Gini index is applied for each Bootstrap. At last, a voting method is used to generate the final prediction.

A feature selection method based on the random forest has been done which is relatively efficient and applicable to high dimensional

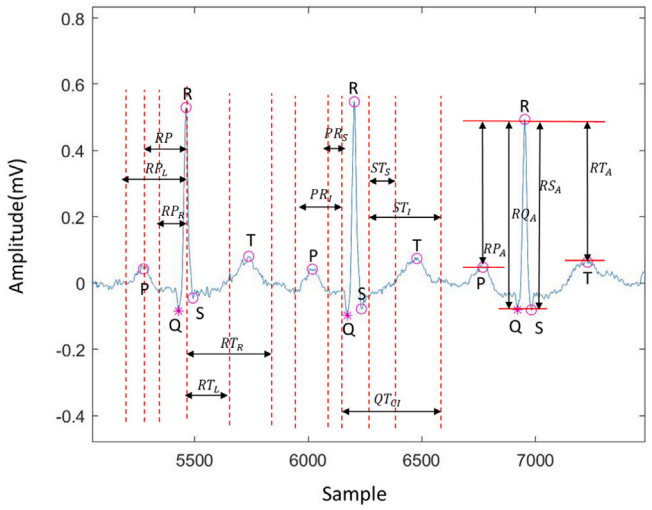


Fig. 6. Features related to the amplitude and interval extracted from a beat.

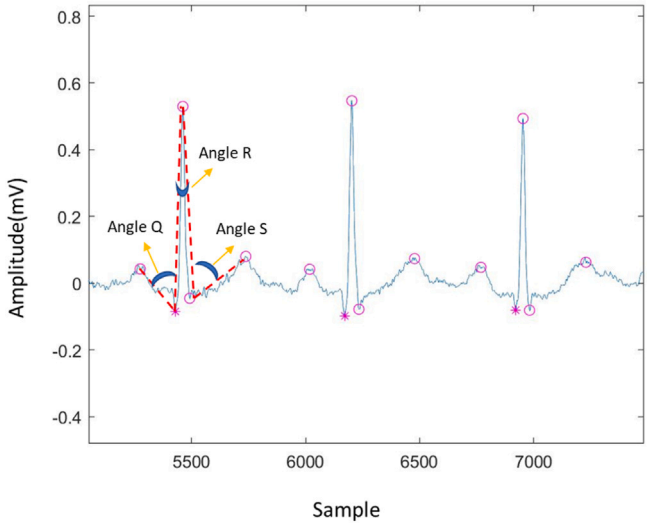


Fig. 7. Features related to the angle extracted from a beat.

data in short time. Also, the random selection of features can improve the performance of the classification [39]. The random forest can determine the importance of features. Thus, it searches how many tree nodes have used that feature to reduce impurity in all trees. It calculates the score for each feature separately in the training phase. Then, the features are selected by a threshold-based approach. This threshold is chosen according to the average importance. The importance of 252 features is shown in Fig. 8.

2.5 Performance evaluation

By studying related works, a series of comparison criteria which are called accuracy (Acc), positive predictive value (PPV), sensitivity (Se), specificity (Sp), and F1 measure (F-score), are common in most of them [40,41].

True Positive (TP): The sick person is correctly classified.

False Positive (FP): A healthy person is misclassified as a patient.

True Negative (TN): A healthy person is correctly classified as healthy.

False Negative (FN): The sick person is misclassified as healthy.

Table 2
List of the extracted features and their representations.

	Features	Representation
Interval features	PR interval	PR_I
	PR segment	PR_S
	QT interval	QT_{CI}
	R_{peak} to T_{offset} segment	RT_R
	R_{peak} to T_{onset} segment	RT_L
	R_{peak} to S_{peak} segment	RS
	R_{peak} to Q_{peak} segment	RQ
	R_{peak} to P_{offset} segment	RP_R
	R_{peak} to T_{onset} segment	RT_L
	R_{peak} to P_{peak} segment	RP
	ST segment	ST_S
	ST interval	ST_I
Angle features	Angle Q	$\angle Q$
	Angle R	$\angle R$
	Angle S	$\angle S$
Amplitude features	RQ amplitude	RQ_A
	RT amplitude	RT_A
	RP amplitude	RP_A
	RS amplitude	RS_A

3 Results

The results are presented in two sections, interpatient and intrapatient. Each of them considers two parts of detection and localization. For an interpatient scheme, training is performed on the records of the particular patient and tested on the new patient's records. So the training and testing data do not overlap. For an intrapatient scheme, shuffled records are divided into testing and training data randomly. Therefore interpatient experiments are close to the real-world applications and are significant from a clinical point of view.

3.1 Interpatient scheme

The interpatient scheme means that the records of people used in the training phase are not used in the test phase. One hundred fifteen people were randomly selected for training and 50 for testing. Then k-fold was applied to the training data with k = 5 and tuned the classifier's parameters. This study presents a new set of morphological features, and astonishing results are obtained in addition to simplicity and comprehensibility.

3.1.1 Results of feature selection

In the feature selection step, one of the random forest advantages was used to select the features. The importance of each feature is obtained based on random forest, and then by setting a threshold, features that have importance higher than the threshold are selected to enter the classifier. The threshold is specified as the average importance of all features. The threshold value is calculated to be 0.78. features with an importance greater than 0.78 are selected. The importance of all features and the threshold is illustrated in Fig. 8A. For MI localization, 34 features are selected. Examining the selected features, it was observed that none are selected from leads I, II, and aVR. Selected features include RS_A, area, QT_{CI}, ST_S, RT_L, S_{angle}, R_{angle}, P_{angle}, RS, RT_A, JH, PR_I, PR_S, RP_A, RP, RQ_A, RS_A, ST_I.

Fig. 8B shows that the importance average was set as the threshold for MI detection. Assuming the threshold of 0.59, 26 features are selected From all 252. The selected features showed that none were selected from leads I, II, III, aVR, and aVL. ST_S, PR_I, RP, and R_{angle} are also not selected.

3.1.2 Results of detection

MI detection is the classification of two classes, MI and HC. After preprocessing and identification of fiducial points, the morphological features were extracted. Then, the features were selected, and the classification model was trained. In MI detection, 115 subjects are

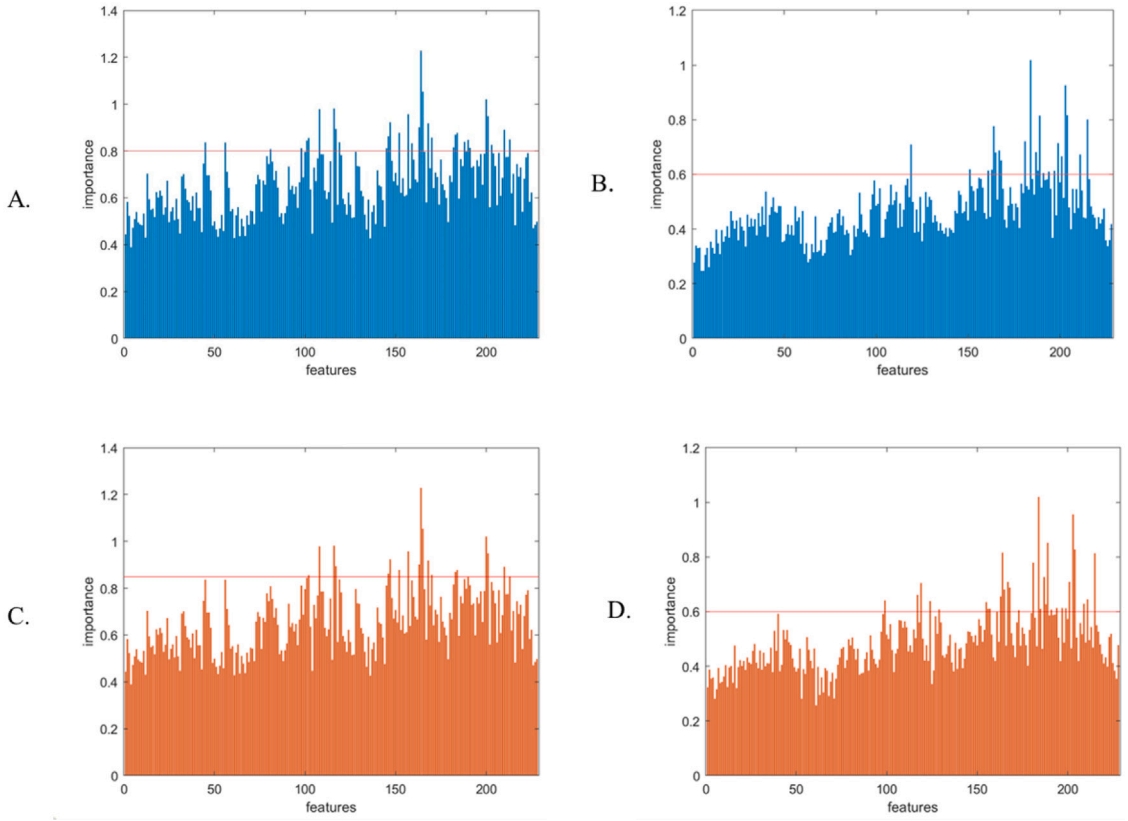


Fig. 8. Importance of 252 features and thresholds specified in the A. inter-patient scheme localization, B. inter-patient scheme detection, C. intra-patient scheme localization, D. intra-patient scheme detection.

Table 3
Confusion matrix and results of MI detection in the inter-patient scheme using chest leads (V1–V6).

	HC	MI	Acc(%)	Sen(%)	Spe(%)	PPV(%)	F-score(%)	MCC	G-mean(%)
HC	3749	673	96.68	99.62	84.78	96.36	97.96	0.89	91.90
MI	68	17830							

Table 4
Confusion matrix and results of MI detection in the inter-patient scheme using 7 leads (avf and V1–V6).

	HC	MI	Acc(%)	Sen(%)	Spe(%)	PPV(%)	F-score(%)	MCC	G-mean(%)
HC	3696	726	96.54	99.74	83.58	96.09	97.88	0.88	91.30
MI	47	17851							

selected to tune classification parameters and select features. After tuning, a model was trained on the 115 subjects. Then the model was applied to the remaining 50 subjects. Fig. 9 shows the corresponding block diagram; since the data in the interpatient scheme is imbalanced, the Precision–Recall curve is used to evaluate the method. Also, the Matthews correlation coefficient (MCC) and Geometric Mean (G-mean) are reported as evaluation criteria in addition to the previous ones. They are defined as follows:

$$MCC = \frac{TP.TN+FP.FN}{\sqrt{(TP+FP).(TP+FN).(TN+FP).(TN+FN)}} \quad (1)$$

$$G-mean = \sqrt{sensitivity \times specificity} \times 100\% \quad (2)$$

Figs. 10 and 11 show the Precision–Recall plot of testing data using 7 leads and chest leads. According to the results, the values of sensitivity and F1-score have been improved. Also, by checking the selected features, it was found that all the selected features, except one, are related to the chest leads. This feature also belong to one lead, meaning that the features are selected from only 7 leads. The confusion matrix and results are reported using chest leads and 7 leads in Tables 3 and 4.

3.1.3 Results of localization

In this section, the results of the proposed algorithm in the inter-patient scheme were reviewed. This scheme is important because of its clinical significance.

As shown in Fig. 9, after tuning the parameters and selecting features, a model was trained on the whole training data. Then the model was applied to the test data. MI localization is more complicated than MI detection because of the 6-class classification problem. Five types of MI are included in the evaluation. From the different types of MI, the ones with the minimum required people are selected: anterior, anterior lateral, anterior septal, inferior, and inferior lateral. One hundred sixty-five subjects are in the dataset, including patients and healthy ones. After feature selection, a random forest was used for classification. The number of trees was selected to be 100 in our setup. Table 5 shows the confusion matrix and results for the classification of each class separately. Table 5 shows that none of the ILMI are detected as HC. Also, only one record is recognized as anterior. In the case of the anterior, only 2 AMIs are considered ILMI. The results show that the algorithm could detect ILMI, HC, and ASMI with better performance. Fig. 12 presents the Precision–recall plot of MI localization.

Table 5
Confusion matrix and results of MI localization in the inter-patient scheme (HC: healthy patients, AMI: anterior MI, ALMI: antero-lateral MI, ASMI: antero-septal MI, IMI: inferior MI, ILMI: infero-lateral MI).

	AMI	ALMI	ASMI	IMI	ILMI	HC	Acc(%)	Sen(%)	Spe(%)	PPV(%)	F-score(%)
AMI	1655	290	733	38	2	117	58.38	58.38	98.13	81.93	68.18
ALMI	82	1569	206	60	57	29	78.33	78.33	93.93	56.00	65.31
ASMI	175	336	3840	232	4	30	83.17	83.17	93.93	78.13	80.57
IMI	4	560	49	4731	401	278	78.55	78.55	97.13	91.00	84.32
ILMI	1	5	6	126	2282	0	94.30	94.30	97.34	81.18	87.25
HC	103	42	81	12	65	4119	93.15	93.15	97.46	90.07	91.58

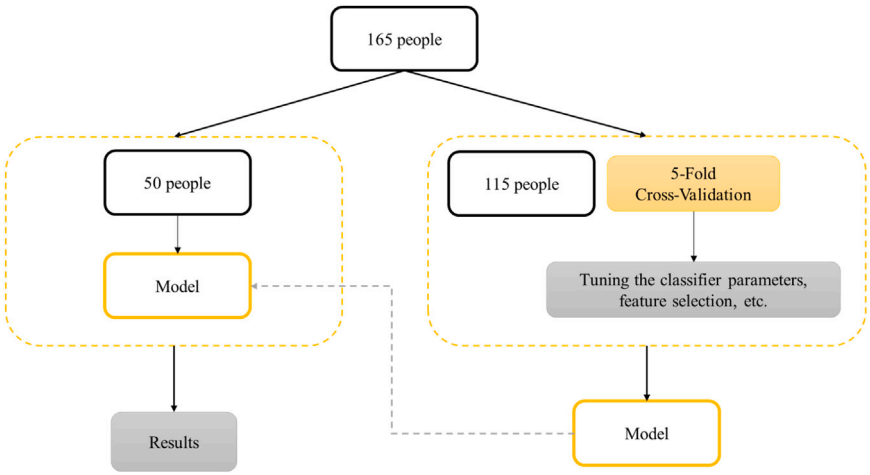


Fig. 9. Block diagram of train and test data for MI detection and localization in inter-patient scheme.

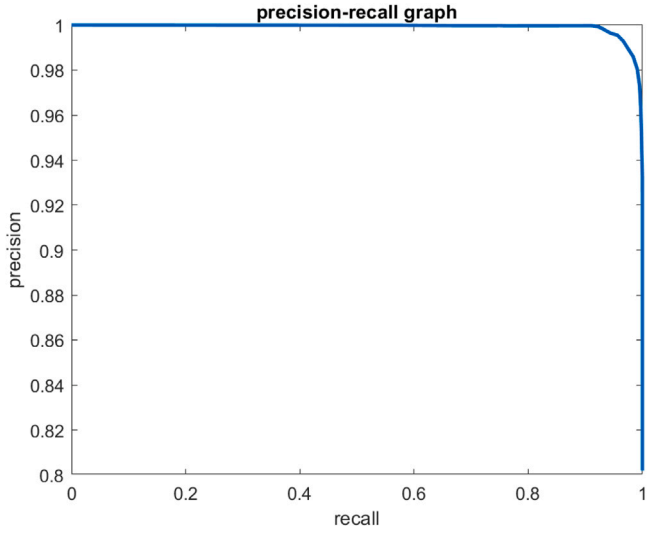


Fig. 10. Precision–recall plot for MI detection in inter-patient scheme using 7 leads.

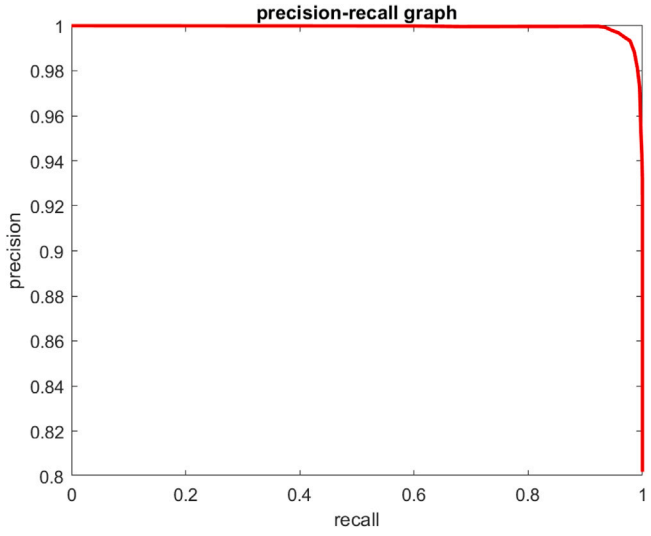


Fig. 11. Precision–recall plot for MI detection in inter-patient scheme using chest leads.

3.2 Intrapatient scheme

In the intrapatient scheme, all records for all people are shuffled. K-fold validation was used with k=5 to evaluate our method. There may be a record from one person in the training and another from the same person in the test. Random forest is used in an intrapatient scheme with a number of trees of 100. The following two sections present the results related to the feature selection and performance of the proposed algorithm in MI detection and localization.

3.2.1 Results of feature selection

For MI localization, the threshold is selected to be 0.85, which is the average importance of the features. Twenty-eight features are selected as the classifier's input, and the results are presented in the next section. In MI detection, the threshold value is 0.6, and 30 features are selected according to that threshold. Fig. 8C and Fig. 8D illustrate the importance of MI localization and detection features in the intrapatient scheme, respectively.

3.2.2 Results of detection and localization

In this section, the results of the proposed algorithm in the intra-patient domain are presented. The main purpose of our study is to examine the interpatient scheme. However, the proposed algorithm is tested in the intrapatient scheme to show that it performs as well as other methods.

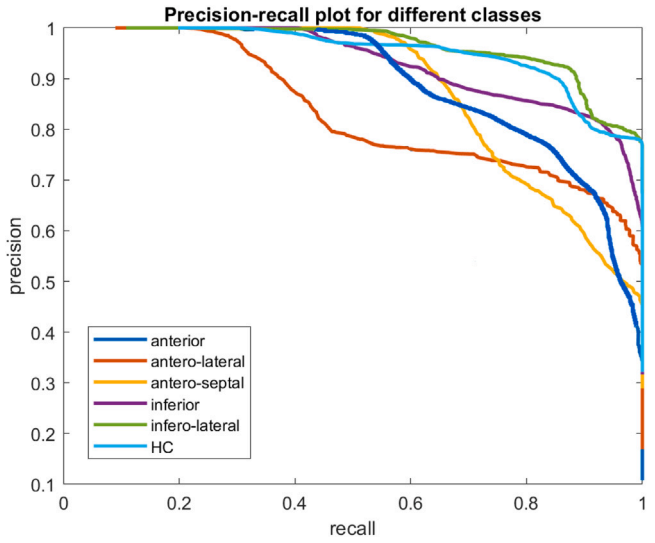


Fig. 12. Precision–recall plot for MI localization in inter-patient scheme.

Table 6
Results of 5-fold cross-validation for MI detection in intra-patient scheme.

Fold	Acc(%)	Sen(%)	Spe(%)	PPV(%)	F-score(%)
1	99.96	99.96	99.96	99.99	99.97
2	99.95	99.97	99.89	99.97	99.97
3	99.95	99.96	99.93	99.98	99.97
4	99.92	99.92	99.89	99.97	99.95
5	99.94	99.93	99.96	99.99	99.96

Table 7
Confusion matrix and results of MI detection in the intra-patient scheme.

	HC	MI	Acc(%)	Sen(%)	Spe(%)	PPV(%)	F-score(%)
HC	14130	10	99.94	99.95	99.93	99.98	99.97
MI	31	59270					

Table 8
Results of 5-fold cross-validation for MI localization in intra-patient scheme.

Fold	Acc(%)	Sen(%)	Spe(%)	PPV(%)	F-score(%)
1	99.83	99.76	99.96	99.83	99.80
2	99.84	99.82	99.97	99.83	99.83
3	99.84	99.78	99.97	99.86	99.82
4	99.83	99.79	99.96	99.84	99.81
5	99.75	99.68	99.95	99.78	99.73
Total	99.81	99.76	99.96	99.82	99.80

After preprocessing, feature extraction, and feature selection, all segments are classified using k-fold cross validation. The confusion matrix and results shown in Table 7 are obtained for MI detection. Accuracy, sensitivity, specificity, precision, and F-score are 99.94%, 99.95%, 99.93%, 99.98%, and 99.97%, respectively. Table 6 shows that the algorithm works well in all folds, and their results are close to each other. The lowest result based on F-score and precision belongs to the fourth fold, which is only 0.02% away from the best result. Therefore, the algorithm in MI detection is not sensitive to training data. According to Table 7, only 31 MI records and 10 HC records are misclassified.

MI localization is a 6-class problem. 12-lead changes need to be monitored. The results for the 5 folds are summarized in Table 8 and the confusion matrix is given in Table 9. Accuracy, sensitivity, specificity, precision, and F-score respectively are 99.81%, 99.76%, 99.96%, 99.82%, and 99.80%. According to Table 9, the F-score and precision values of ALMI are lower than other types of MI. On the other hand, none of the records of AMI as ILMI, ALMI as AMI, and ILMI as IMI have been identified.

4 Discussion

Numerous works have been done on MI detection and localization in the intrapatient scheme. There is less focus on MI localization in the interpatient scheme. Table 10 compares the results with similar articles.

The paper [30] is the first paper on MI localization using 12-lead ECG in the interpatient scheme. It uses a deep learning algorithm and needs more data to provide better results and has stated that the low number of patients with MI types is the reason for the low result obtained for MI localization in the interpatient scheme. While the proposed algorithm achieved the desired results using the same database. Article [30] has also shown that the wrong labels will be obtained and are not interpretable using one or more leads instead of 12. The effect of various leads has been investigated in the proposed method for interpatient MI detection. In this work, in addition to the low complexity of the proposed algorithm and the use of features that are interpretable to experts, better results have been achieved in comparison to the related works.

The method presented in the article [43] is computationally complex, requires high training time, and is memory inefficient. This method also has a complex structure and extensive parameters, thus requiring optimization. It is challenging for cases where processing power is limited, especially in portable devices. In the proposed method, the problem is addressed. Using this package of features together has worked very efficiently, and the ECG signal changes caused by MI are accurately detected. In the article [43], effective lead selection, which has a greater impact on MI detection and localization, has not been made; thus, it requires comparative experiments and analysis. In the proposed method, this weakness has been eliminated. In the MI detection on the interpatient scheme, 6 chest leads, and in the inferior MI diagnosis section, only one lead was used. Also, the algorithm presented in [43] requires higher training data, while the proposed method has achieved good results with less training data.

The paper [15] used the frequency domain and wavelet transform properties, though it failed to show the differences caused by MI in the ECG signal well.

The proposed method uses a random forest classifier, an ensemble learning method. It has been very popular in recent articles due to its versatility, which means it can be used in classification and regression. It is also possible to easily determine the importance of input features. One of the big problems in machine learning is overfitting, which rarely occurs in a random forest classifier if there are enough trees in the forest. In the article [42], MI has been diagnosed using the MFB-CBRNN neural network. Two studies [30,43], in the interpatient scheme, used the neural network and achieved good accuracy in MI detection. The proposed method extracts describable features to diagnose the MI disease. This algorithm could achieve better results with lower computational complexity than preview works. Article [28] based on deep learning and knowledge graph was proposed. It was different from deep learning method belonging to black box. Articles [28,30,43] have not performed well in MI localization. Article [29] proposed a multi-task channel attention network (MCA-net). In this paper, MCAnet is prone to confusion when differentiating the relevant labels and needs more data.

Table 10 shows that the results obtained in the proposed method, compared to the article [43], in the MI detection of the interpatient scheme have been improved by about 18%. Thus, it can be claimed that the proposed algorithm works better than the recent studies. In our method, a new package of morphological features has been used, which has interpretability, and the algorithm analyzes different leads of ECG. In addition, experiments have been performed in the interpatient scheme, which is important from a clinical point of view.

Table 9
Confusion matrix and results of MI localization in the intra-patient scheme (HC: healthy patients, AMI: anterior MI, ALMI: antero-lateral MI, ASMI: antero-septal MI, IMI: inferior MI, ILMI: infero-lateral MI).

	AMI	ALMI	ASMI	IMI	ILMI	HC	Acc(%)	Sen(%)	Spe(%)	PPV(%)	F-score(%)
AMI	6958	4	3	3	0	5	99.78	99.78	99.98	99.78	99.78
ALMI	0	7877	2	4	3	2	99.86	99.86	99.90	99.18	99.52
ASMI	10	19	16730	8	3	1	99.76	99.76	99.98	99.95	99.85
IMI	2	20	2	16830	3	3	99.82	99.82	99.97	99.89	99.86
ILMI	1	12	1	0	10760	0	99.87	99.87	99.98	99.88	99.87
HC	2	10	1	3	4	14150	99.86	99.86	99.98	99.92	99.89

Table 10
Compare the results with similar articles on PTB Dataset.

Author	Number of subjects or records	Features	Classifier	Inter-patient		Intra-patient	
				Detection(%)	Localization(%)	Detection(%)	Localization(%)
[15]	148 MI subjects, 52 HC subjects	Features such as energy entropy based on MODWPT and morphological features	SVM-RBF	Acc=92.69, Sen=80.96, F-score=83.26	–	Acc=99.75, Sen=99.37, F-score=99.54	–
[42]	148 MI subjects, 52 HC subjects	Generated features by the neural network	MFB-CBRNN	Acc=93.08, Sen=94.42, Spe=86.29	–	Acc=99.90, Sen=99.97, Spe=99.54	–
[30]	113 MI subjects, 52 HC subjects	Generated features by the neural network	ML-ResNet	Acc=95.49, Sen=94.85, Spe=97.37, F-score=96.92	Acc=55.74, Sen=47.58, Spe=55.37, F-score=47.94	Acc=99.92, Sen=99.98, Spe=99.77, F-score=99.94	Acc=99.72, Sen=99.63, Spe=99.72, F-score=99.67
[43]	113 MI subjects, 52 HC subjects	Generated features by the neural network	MLA-CNN-BiGRU	Acc=96.50, Sen=97.10, Spe=93.34	Acc=62.94, Sen=63.97, Spe=63.00	Acc=99.93, Sen=99.99, Spe=99.63	Acc=99.11, Sen=99.02, Spe=99.10
[28]	17212 MI records, 6945 HC records	–	Knowledge graph; beat morphology recognition; production rules	Acc=97.61, Sen=97.23, Spe=98.73	Acc=74.34, Sen=70.31, Spe=79.04	–	–
[29]	113 MI subjects, 52 HC subjects	–	MCA-net	Acc=95.57, Sen=98.01, Spe=86.01	Acc=68.90, Sen=55.35, Spe=67.96	–	–
Proposed method	113 MI subjects, 52 HC subjects	Morphological features	Random Forest	Acc=96.54, Sen=99.74, Spe=83.58, PPV=96.09, F-score=97.88	Acc=80.98, Sen=80.98, Spe=96.32, PPV=79.72, F-score=79.53	Acc=99.94, Sen=99.95, Spe=99.93, PPV=99.98, F-score=99.97	Acc=99.81, Sen=99.76, Spe=99.96, PPV=99.82, F-score=99.80

5 Conclusion

Using the ECG changes caused by MI, this study presents a simple and practical algorithm for MI detection and localization. The PTB database was used to evaluate our method. By extracting simple morphological features, fair results are achieved in both interpatient and intrapatient schemes. Identifying the fiducial points of the ECG signal is an important contribution to the extraction of features. A random forest classifier was used with 100 trees and a K-fold with k = 5. The results indicate that the proposed method works superior to the recent works on MI detection and localization. The results show that studying MI characteristics and awareness of its physiological distortions on ECG signals will help engineers to improve MI detection based on ECG signals.

CRediT authorship contribution statement

Sahar Ramezani Moghadam: Conceptualization, Methodology, Software, Data process, Writing – original draft, Writing – review & editing. **Babak Mohammadzadeh Asl:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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